Seasonal Median Imputation Report

Project Information

Item	Value
Project Name	EnergyPulse - Smart Energy Consumption Analytics
Module	AID-3 Scalable Chunk Pipeline & Preprocessing
Deliverable	Seasonal Median Imputation
Author	Salwa Gamal Ahmed Agha
Version	1.0

1. Objective

Handle missing values while preserving dataset records using Seasonal Median Imputation.

2. Input Dataset Status

Missing values existed as both ? symbols and NaN values across key energy features.

3. Data Preparation

Missing values were standardized, numeric columns converted to float64, and a unified Datetime column was created.

4. Seasonal Feature Engineering

Hour, Month, and DayOfWeek features were extracted from Datetime.

5. Seasonal Median Imputation Methodology

Hourly median values were calculated and used to replace missing values within the corresponding hour group.

6. Validation Before Imputation

Each affected numeric column contained 8 missing values within the 100,000-row validation sample.

7. Validation After Imputation

All missing values were successfully replaced and remaining missing values were reduced to zero.

8. Data Integrity Verification

Row count remained 100,000 before and after imputation.

9. Deliverables Produced

Standardization, datatype conversion, datetime construction, seasonal feature extraction, imputation logic, and validation were completed.

10. Engineering Benefits

Improved data quality, preserved temporal patterns, and increased readiness for forecasting and anomaly detection.

11. Conclusion

Seasonal Median Imputation was validated successfully and prepared the dataset for downstream processing.

Validation Results

Column	Before	After
Global_active_power	8	0
Global_reactive_power	8	0
Voltage	8	0
Global_intensity	8	0
Sub_metering_1	8	0
Sub_metering_2	8	0
Sub_metering_3	8	0